

ANNA BURGESS

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EDUCATION

Harvard College | Cambridge, MA

Exp. May 2027

Bachelor of Science in Mechanical Engineering, Minor in Mathematical Sciences, GPA: 3.89

Relevant Coursework: Computer-Aided Machine Design, Mechanical Systems, Solid and Computational Mechanics, Robotics

WORK EXPERIENCE

Head Course Assistant | Introduction to Computer-Aided Machine Design, Harvard

Oct. 2024 – Present

- Instruct 1–2 teams of 4–6 students in machining and the design and fabrication of a competition robot
- Lead weekly office hours for 10–20 students covering statics, G-code, SolidWorks, GD&T, and manufacturing processes
- Grade technical assignments and SolidWorks designs for classes of 40–60 students, providing individualized feedback

Mechanical Engineering Intern | Flex

May 2026 – Aug. 2026

- Developed 26 production fixtures to reduce cycle time, component damage, and assembly variation
- Designed machined product nest for 13-station automated line to withstand 65°C processing without deformation
- Produced 178 SolidWorks parts/assemblies and 25+ peer-reviewed drawings with dimensional tolerancing/stack-ups
- Designed alignment and handling fixtures to resolve operator-reported assembly and quality issues

Maker Fellow | REEF Makerspace, Harvard

Aug. 2025 – May 2026

- Independently managed makerspace operations; assisted students with laser cutting, soldering, and 3D printing
- Designed example projects and wrote documentation to expand use of an underutilized desktop milling machine

Undergraduate Research Assistant | Ability Lab, Harvard

Aug. 2025 – May 2026

- Redesigned visual-perturbation headset, replacing failure-prone acrylic assemblies with lightweight 3D-printed PLA
- Designed heat-set insert, friction-fit mirror, and servo interfaces to improve durability, assembly, and serviceability

Research Intern | Mechatronics and Haptic Interfaces Lab, Rice University

May 2025 – July 2025

- Designed force-sensitive resistor (FSR) housings for a center-of-pressure sensing insole and force-quantifying scoliosis brace
- Developed equation-driven, multi-material CAD and validated FSR designs through 125 Instron trials to 200 N
- Validated scoliosis-brace sensors and fabricated 10 units for handoff to Baylor Scott & White for clinical testing
- Identified wire-management and conductive-TPU limitations affecting sensor consistency and iterated insole design

ENGINEERING PROJECTS & LEADERSHIP

Chief Engineer and Project Manager | Harvard Undergraduate Aerospace Collective - Rocketry

Oct. 2025 – Present

- Founded Harvard's 10,000-ft IREC rocketry program and led 5 members through NAR Level 1 certification
- Led development of a liquid bipropellant regeneratively cooled engine with a metal 3D-printed combustion chamber
- Coordinated four technical subteams through assembly and static fire of a 250-lbf, 5,000-N·s bipropellant rocket
- Initiated a 3–5 km LoRa telemetry project and designed avionics housings, flight-tested aboard micro-G liquid rocket
- Managed team budget, project schedule, transportation logistics, onboarding, and outreach alongside technical leadership

Machining Lead | Harvard Undergraduate Robotics Club

Oct. 2024 – Present

- Machined precision power-transmission components to ± 0.003 -in tolerances for a University Rover Challenge Mars rover
- Selected manufacturing processes and redesigned components to reduce machining complexity and setup time
- Co-led a 40-member mechanical team, allocating work across arm, chassis, suspension, and science module

Structures Lead | Harvard Undergraduate Rocket and Propulsion Group

Oct. 2024 – Oct. 2025

- Performed SolidWorks FEA on rocket bulkheads under high loads, verifying structural integrity
- Designed, analyzed, and fabricated an 800-lbf rocket engine static test stand with a 2× factor of safety

TECHNICAL SKILLS

Manufacturing: Manual/CNC Turning and Milling, MIG/TIG Welding, 3D Printing, Laser/Waterjet Cutting, Soldering

CAD/CAE/CAM: SolidWorks, Fusion 360, Abaqus, COMSOL

Programming: Python, MATLAB, Arduino | **CNC:** G-code, ProtoTRAK Conversational

Certifications: CSWA, NAR Level 1 High Power Rocketry